

IEEE Standard for General Requirements for Identity Framework for Metaverse

IEEE Consumer Technology Society

Developed by the
Digital Finance and Economy Standards Committee

IEEE Std 3812.1™-2023

IEEE Standard for General Requirements for Identity Framework for Metaverse

Developed by the

Digital Finance and Economy Standards Committee
of the
IEEE Consumer Technology Society

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IEEE SA Standards Board

Abstract: The requirements of an identity framework for metaverse is defined in this standard. An identity framework for use across different metaverse systems is provided. Furthermore, the relevance between real world and virtual world entities is recognized. Business logic, operational procedures, and authentication programs is covered in this standard. Also, terminologies, a basic architectural framework, and key indicators are defined.

Keywords: Identity, identity framework, identity framework for metaverse, IEEE 3812.1™, metaverse

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Introduction

This introduction is not part of IEEE Std 3812.1-2023, IEEE Standard for General Requirements for Identity Framework for Metaverse.

IEEE 3812 standards provide general identity framework for multiple systems. Within the context of the IEEE 3812 family of standards for identity framework, this standard defines the requirement of an identity framework for metaverse. This framework is designed to address the challenges of interoperability, scalability, security, and privacy in multiple metaverse systems. The framework includes a range of functions, including identifier functions, identity functions, interoperable identity functions, and identity tool and maintenance functions. Each function is designed to perform specific tasks, such as managing identifiers, allocating and abstracting resources, managing identity relationships, managing credentials and claims, and security and regulation compliance. By adhering to these principles, the identity framework can facilitate secure and seamless interactions between users, smart devices, AIs/robots, data instances in multiple applications, and virtual environments, which can help on creating a more immersive and trustworthy metaverse, as well as an open-loop multiverse ecosystem.

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IEEE Standard for General Requirements for Identity Framework for Metaverse

1. Overview

The metaverse includes both extended reality and digital twins, defined by various systems including Internet of Things, Information Technology, and the Internet. Each system has an internal entity definition for an end user, smart device, artificial intelligence (AI)/robot, sensor, and data instance. Each entity in the metaverse needs to be recognized across the multiple systems within the defined identity framework.

1.1 Scope

This standard defines the requirements of an identity framework for metaverse.

This standard provides an identity framework for use across different metaverse systems. Furthermore, the standard helps to recognize the relevance between real world and virtual world entities. The standard covers business logic, operational procedures, and authentication programs. Also, the standard defines terminologies, a basic architectural framework, and key indicators.

1.2 Purpose

This project provides an identity framework to satisfy the requirement of entities to interoperate/interact cross multiple metaverse systems. It includes identity recognition in multiple systems, identity profile management and presentation in different systems, and id-based access control for data interoperability cross systems (authentication, authorization, and accounting).

1.3 Word usage

The word *shall* indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (shall equals is required to).^{6,7}

The word *should* indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (should equals is recommended that).

⁶The use of the word *must* is deprecated and cannot be used when stating mandatory requirements, *must* is used only to describe unavoidable situations.

⁷The use of *will* is deprecated and cannot be used when stating mandatory requirements, *will* is only used in statements of fact.

The word *may* is used to indicate a course of action permissible within the limits of the standard (may equals is permitted to).

The word *can* is used for statements of possibility and capability, whether material, physical, or causal (can equals is able to).

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

There are no normative references in this standard.

3. Definitions, acronyms, and abbreviations

3.1 Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.⁸

anonymous credential: An anonymous credential is transformed from certain credential by the user to contain only some of the original information and applies mathematical functions to prove attribute properties without revealing the attribute itself.

decentralized identifier: Decentralized identifiers (DIDs) are a type of identifier that enables disintermediate verification to digital identity, the verification issuer will not take part in the verifying process.

NOTE—A DID specification can refer to “decentralized identifier” from W3C [B1].⁹

identity presentation: Data derived from one or more identity resources from different metaverse applications, which can be shared as a specific entity to a metaverse, and can be verified.

identity profile: An identity profile is an identity presentation based on an identity profile template.

identity profile template: Identity profile template is a set of data attributes that are used to verify the identity of an entity in certain applications, such as login credentials, IP address, geolocation, and behavioral patterns.

metaverse: Metaverse refers to a kind of experience in which the outside world is perceived by the users (human or non-human) as being a universe that is built upon digital technologies as a different universe (“Virtual Reality”), a digital extension of our current universe (“Augmented Reality”), or a digital counterpart of our current universe (“Digital Twin”). In a broad sense, metaverse is the advanced stage and long-term vision of Digital Transformation.

NOTE—The definition of “metaverse” is from [B2].^{10,11}

⁸*IEEE Standards Dictionary Online* is available at: <http://dictionary.ieee.org>. An IEEE Account is required for access to the dictionary, and one can be created at no charge on the dictionary sign-in page.

⁹Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

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¹¹The numbers in brackets correspond to those of the bibliography in Annex A.

verifiable claim: Verifiable claims are the actual pieces of information or attributes about a subject that can be cryptographically verified for its authenticity and accuracy. Verifiable claims are part of the broader concept of verifiable credentials.

verifiable credential: A verifiable credential is a digital representation of information that can be cryptographically verified for authenticity and accuracy.

NOTE—The data model of verifiable credential can refer to “Verifiable Credentials Data Model” from W3C [B3].

3.2 Acronyms and abbreviations

AAA	Authentication, Authorization, and Accounting
AC	Anonymous Credential
AI	Artificial Intelligence
DID	Decentralized Identifier
VC	Verifiable Credential

4. Identity framework for Metaverse

4.1 Overview of identity framework for Metaverse

An identity framework refers to a set of guidelines, standards, and protocols for managing digital identities. It provides a common structure for creating, storing, and exchanging identity information and supports interoperability and security for digital identity management.

The metaverse refers to a virtual shared space that provides immersive digital experiences and enables users to interact with each other and with virtual objects in a persistent and seamless manner.

The “identity framework for metaverse” refers to a set of guidelines, standards, and protocols that govern the management and representation of digital identities within virtual shared spaces or metaverse environments.

An identity framework for metaverse aims to provide a secure and interoperable way of representing digital identities in virtual shared spaces. It provides a common structure for creating, storing, and exchanging identity information, and supports the interoperability of different metaverse environments and platforms.

A view of identity framework for metaverse is described in [Figure 1](#).

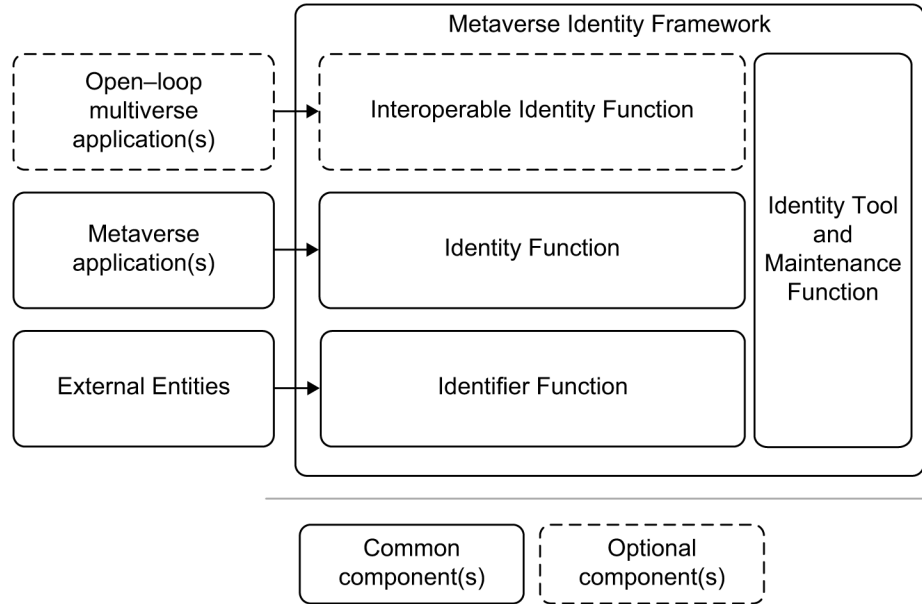


Figure 1—A view of identity framework for metaverse

The following are four types of functions in the identity framework which work with other systems within the metaverse, such as virtual reality (VR) or augmented reality (AR) devices:

- a) Identifier functions
- b) Identity functions
- c) Interoperable identity functions
- d) Identity tool and maintenance functions.

4.2 Components of identity framework for metaverse

The components of identity framework for the metaverse are as follows:

- **Identifier:** an identifier works with external entities by enabling secure and interoperable exchange of identity information between systems. The use of a common identifier provides for interoperability and enables external entities to access relevant identity information linked with the identifier of an entity.
- **Identity:** an identity in a metaverse can work with metaverse applications by using a unique identifier, verifiable claims, consented data sharing, and a good user experience to help enable access to content, services, and experiences within the metaverse. The identity system plays a crucial role in enabling these interactions, and in privacy and security.
- **Interoperable identity:** an interoperable identity in a cross-multiverse environment enables seamless and secure communication between different metaverse applications and systems. The interoperable identity system shall have interoperable identifiers, support verifiable claims, and allow consented data sharing with specified data profiles across multiple metaverses. It shall also access content, services, and experiences in a seamless and secure manner.
- **Identity Tool:** identity tools work by providing different functionalities that support and enhance the management of digital identities in the metaverse(s). These tools should be designed with a focus on user experience so that users can easily use them and access the benefits that they offer in a seamless and secure manner.

5. Requirements for an identity framework for metaverse

An identity framework for metaverse should provide a secure and user-friendly platform for identity management in the metaverse, supporting the growth and development of this virtual world.

The metaverse includes both extended reality and digital twins, defined by various systems, including Internet of Things, Information Technology, and the Internet. Each system has an internal entity definition for an end user, smart device, artificial intelligence (AI)/robot, sensor, and data instance. Each entity in the metaverse needs to be recognized across the multiple systems within the defined identity framework.

To support identity framework for metaverse, many aspects should be considered.

5.1 Functional requirements for an architectural identity framework for metaverse

The functional requirements for an architectural identity framework for metaverse are shown in [Figure 2](#).

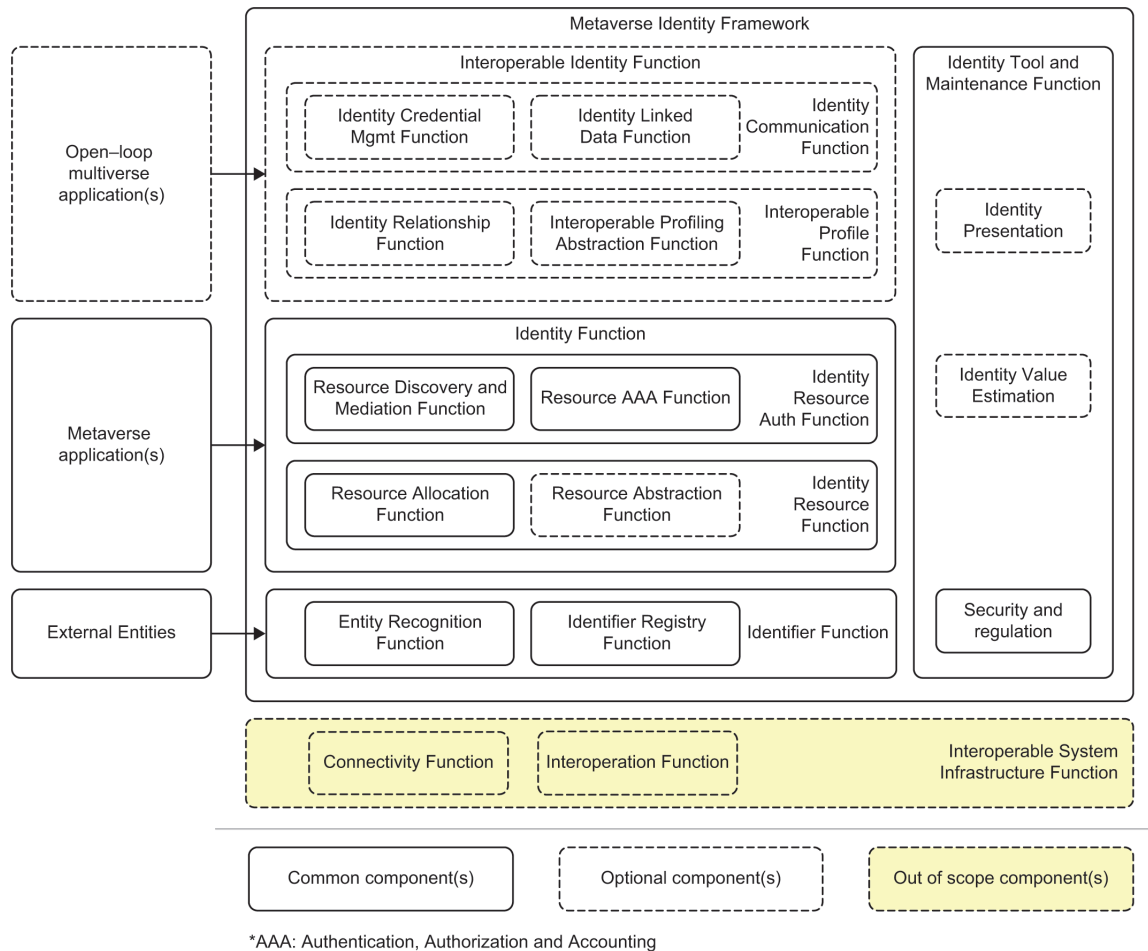


Figure 2—Functional requirements of identity framework for metaverse

5.1.1 Identifier functions

An identity framework for metaverse shall support identifier functions to identify external entities from different systems and shall enable identifier functions to communicate with identity functions (see 5.2). As a result, a metaverse identifier shall enable the following features:

5.1.1.1 Unique identifier

An identifier works with external entities by enabling secure and interoperable exchange of identity information between systems. An identifier acts as a unique identifier for an entity within the identity system, and it shall enable the system to link the entity with its attributes and data.

5.1.1.2 Interaction

When an external entity wants to interact with the identity system, it shall use the identifier of the entity it wants to interact with. The external entity may then use APIs, protocols, or other communication channels to access the identity information linked with the identifier. The identity system may then use the identifier to retrieve the relevant identity information, such as personal information, verifiable claims, and consent records, and provide it to the external entity.

5.1.1.3 Interoperability and secure exchange

The use of a common identifier should provide for interoperability and should enable secure exchange of identity information between systems. For example, if the external entity is an authorized banking system, it may use the identifier of a person to securely retrieve their financial information, or if it is an authorized healthcare system, it may use the identifier to securely retrieve the person's health records.

5.1.2 Identity functions

An identity framework for metaverse shall support identity functions to work with metaverse applications to enable access to content, services, and experiences within the metaverse and shall enable identity functions to manage external entities via identifiers (see 5.3). As a result, a metaverse identity shall enable the following features:

5.1.2.1 Unique identifying

The metaverse identity functions shall use a unique identifier to identify a person or entity within the metaverse. This identifier can then be used by metaverse applications to access the relevant identity information.

5.1.2.2 Verifiable access control

The metaverse identity functions shall use certain digital statements of fact that can be independently verified and trusted in all systems of a metaverse application. They shall provide a secure and reliable way of verifying information and establishing trust between entities, and they shall be able to provide the necessary permissions to access certain content or services within the metaverse.

5.1.2.3 Consented data sharing

The metaverse identity functions shall enable users to manage and control their personal data, and to consent to data sharing with metaverse applications, consistent with all applicable laws and regulations. Privacy and data protection should enable metaverse applications to access the data they need to provide services and experiences to users, consistent with all such legal and regulatory requirements.

5.1.2.4 User experience

The metaverse identity functions may provide users with a simple and intuitive interface for managing their identity information and consenting to data sharing, and it can provide metaverse applications with the necessary identity information to create personalized experiences for users.

5.1.3 Interoperable identity functions

An identity framework for metaverse may support interoperable identity functions to enable users to use the same identity across multiple metaverses, and to access content, services, and experiences in a seamless and secure manner with specified identity profiles for different metaverses (see 5.4). As a result, an interoperable metaverse identity should enable the following features:

5.1.3.1 Interoperable identifier

The metaverse interoperable identity functions shall use identifiers that are interoperable across multiple metaverses. This could be achieved using common standards, such as decentralized identifiers (DIDs) or secure digital certificates.

5.1.3.2 Verifiable claims

The metaverse interoperable identity functions should support verifiable claims to establish trusted communication channel for metaverses, which can be used to verify user attributes, such as age, education, employment history, or other attributes, across multiple metaverses.

5.1.3.3 Interoperable data profile sharing

The metaverse interoperable identity functions shall allow users to manage and control their personal data, and to consent to data sharing with metaverse applications using different identity profiles, consistent with all applicable laws and regulations. Privacy and data protection should enable metaverse applications to access the data they need to provide services and experiences to users, consistent with all such legal and regulatory requirements.

5.1.3.4 Trust interoperable infrastructure

If the metaverse interoperable identity functions are enabled in metaverse identity system, interoperable system infrastructure functions may be used to verify the trust between multiple metaverses, and to setup trust communication channels or safe storage for multiverse connectivity and interoperations.

5.1.4 Identity tool and maintenance functions

An identity framework for metaverse should support identity tool and maintenance functions to support and enhance the management of digital identities in the metaverse (see 5.5).

5.2 Requirements for generic identifier functions for metaverse identity framework

Generic identifier functions of metaverse identity framework provide the services of identifier registration and identifier-based entity recognition.

Identifier registry function is a component of the identifier function in an identity framework. It shall be able to handle the process of registering and maintaining a unique identifier for each entity, such as individuals, organizations, or smart devices, AIs/robots, and sensors within a metaverse. Identifier registry function shall meet the following requirements:

- a) The identifier registry function shall be responsible for assigning a unique identifier to each entity and for maintaining the accuracy and validity of these identifiers over time.
- b) The identifier registry function shall define the data model, guidelines, and protocols of an identifier for external systems to follow to generate and register identifiers for external entities. Especially, for identifiers that are assigned to entities with no sovereignty but operational authority, like smart devices, AIs/robots, or sensors, the identifier may contain verification information that links to the entity owners, or entity controllers.
- c) The identifier registry function shall be designed so that each entity has a unique and persistent identifier that can be used for a variety of purposes, such as verifying the entity's identity, tracking the entity's activities within the metaverse, or establishing relationships with other entities.
- d) The identifier registry function shall be designed so that identifiers are secure and can be relied upon by other entities within the metaverse. It is important that the identifier registry function is implemented in a secure and privacy-preserving manner to help prevent identity theft, fraud, or other forms of abuse. Especially, for identifiers that are designed to be held by the entity themselves, identifier revoke or expire operating to against malicious actions to the identifier shall be considered, identifier revoke or expire operator may be linked, or assigned to the identifier.
- e) The identifier registry function may be maintained by a trusted third party, such as a government or a consortium. When the identity information comes from different metaverses, or different systems to serve metaverse applications, the identifier registry function should be maintained by a trusted third party.

Entity recognition function is a component of an identifier function in an identity framework. It shall be able to recognize and differentiate between different entities, such as individuals, organizations, or smart devices, AIs/robots, sensors, in the metaverse. The entity recognition function shall meet the following requirements:

- a) The entity recognition function shall define a set of guidelines, standards, and protocols for multiple authentication forms of unique identifier values, such as digital certificates, biometrics, or other forms of authentication. It shall provide support for those authentication form in different systems that an entity is using.
- b) The entity recognition function shall associate the correct digital identity with each entity, and verify that the right level of access and permissions are granted to each entity. Especially, for entities with no sovereignty but operational authority, like smart devices, AIs/robots, or sensors, and their identifiers that contain verification information to the entity owners, or entity controllers, the function may be crucial for verifying that the right level of access and permissions are granted to each entity owner, or entity controller.

5.3 Requirements for generic identity functions for metaverse identity framework

Generic identity functions of metaverse identity framework provide the services of identity resource management and identity-based resource authentication management.

The identity resource function in the identity function of an identity framework should be responsible for managing resources related to digital identities. This function may include resource allocation function, which determines how identity-related resources are allocated and assigned; and resource abstraction function, which abstracts the underlying resources into a common, usable form to support cross system resource mediation essentially. The identity resource function shall meet the following requirements:

- The resource abstraction function should define and provide supporting to the process of abstracting and hiding the underlying complexity of the resources used by the identity framework.

- a) The resource abstraction function helps to simplify the process of accessing and using the resources required for the operation of the identity framework and enables entities to interact with these resources in a more intuitive and user-friendly manner. By abstracting the underlying complexity of the resources, the resource abstraction function may help to reduce the barriers to entry for entities and make it easier for them to use the identity framework and its associated services.
- b) The resource abstraction function may support the security and privacy of the identity framework, by abstracting away the underlying technical details of the resources and making it more difficult for malicious actors to attack or exploit them.
- The resource allocation function shall define and provide supporting to the process of allocating and managing the resources, such as computing power, storage, bandwidth, or other assets, required to support the operation of the identity framework and its associated services.
 - a) The resource allocation function should verify that the identity framework has the resources it needs to function effectively and efficiently, and that these resources are used in a fair and equitable manner.
 - b) Resource allocation may be performed in a variety of ways, such as through a marketplace model, where entities bid for and purchase resources, or through a more centralized allocation model, where resources are assigned based on some predefined criteria.
 - c) The resource allocation function can help the metaverse support a high volume of digital interactions and transactions and can scale to meet the needs of a growing number of entities.

The identity-based resource authentication function in the identity function of an identity framework shall define and provide supporting to the set of processes and mechanisms that enable the secure and authorized access to an individual's identity information and associated resources. This function shall include resource AAA (Authentication, Authorization, and Accounting) function, which enables the verification of the individual's identity, authorization for access to specific resources, and enforcement of access control policies; and resource discovery and mediation function, which enables the discovery and mediation of identity-related resources across different systems and applications.

The goal of this function is that only authorized parties should have access to the individual's identity information and related resources, and that the access should be secure and in line with the individual's privacy preferences. The identity-based resource authentication function shall meet the following requirements:

- The resource discovery and mediation function shall define and provide supporting to the process of discovering, locating, and mediating access to the resources required for the operation of the identity framework.
 - a) The resource discovery and mediation function helps entities to locate and access the resources they need, and use the resources in an appropriate and secure manner. The resource discovery and mediation function may be responsible for tasks such as authenticating entities, validating their access rights, and mediating access to the resources based on these access rights.
 - b) The resource discovery and mediation function shall support properly controlled access to the resources, that is only granted to authorized entities. This can help to reduce unauthorized access to the resources and protect the privacy of the entities using the identity framework.
 - c) The resource discovery and mediation function may define and provide supporting to identity data mediation cross different systems, including syntactic mediation and semantic mediation.
- The identity resource AAA function shall define and provide supporting to the process of verifying the identity of an entity and granting or denying access to specific resources based on the authorization policies defined. This function helps verify that only authorized entities have access to the protected resources and provides a mechanism to track and audit the usage of these resources.

- a) The identity resource AAA function shall provide mechanisms for verifying the identity of entities trying to access the resources. This includes user authentication and device authentication.
- b) The identity resource AAA function shall provide mechanisms for defining the access rights of entities to specific resources and for enforcing those rights.
- c) The identity resource AAA function shall provide mechanisms for tracking the usage of resources by entities. This includes logging of resource access, generation of usage reports, and billing.
- d) In addition to these core requirements, the identity resource AAA function shall also comply with all applicable requirements for data and transaction security, and to protect the privacy of the users.

5.4 Requirements for interoperable identity functions for metaverse identity framework

Interoperable identity functions of metaverse identity framework may provide the services of interoperable identity profile management and identity communication management.

An interoperable identity profile is a digital representation of a person's identity that can be used in a consistent and standardized manner across multiple systems, applications, and services. It contains relevant information about the entity and its identity-based behavior, such as name, address, contact information, driving history, and other details that are needed to identify and verify their identity.

In an identity framework for multiple metaverses, identity data should be accumulated in the identity framework from all metaverse applications and be generating interoperable identity profile per request of each metaverse application.

The interoperable profile function of an identity framework for metaverse should verify that identity data is interoperable, secure, and private, and can be used across multiple systems, applications, and services. It should support the creation, management, and use of interoperable identity profiles upon overall identity data across different metaverses, as follows:

- The interoperable identity profiling abstraction function shall be able to abstract and normalize identity data from different sources into a common, interoperable format. This function shall also generate identity profile to meet the interoperable data requirement for different metaverses. This function may provide a common interface for accessing and processing identity data, helping to keep identity data secure, private, and interoperable. The interoperable profile function shall meet the following requirements.
 - a) The interoperable identity profiling abstraction function shall allow entities to aggregate and manage identity data from different sources or different metaverse applications, such as social media profiles, government-issued identity documents, and other identity management systems. The interoperable identity profiling abstraction function provides a common interface for accessing and processing identity data, allowing entities to interact with identity data in a consistent, interoperable manner.
 - b) The interoperable identity profiling abstraction function should allow any open-loop metaverse application to provide its identity data format, require identity profile from the identity framework, and accept the identity based on the identity profile output.
 - c) The interoperable identity profiling abstraction function should also help to keep identity data secure and private. For example, it can provide privacy controls to manage access to identity data, or encrypt identity data to help prevent unauthorized access.
- The identity relationship function shall enable entities to establish trust-based relationships with each other, facilitating secure and interoperable communication and exchange of information across

multiple metaverse systems. This function shall also be able to provide the identity data management to different identities in different metaverse systems when the data is generated in any actions within relationship interaction scenarios.

- a) The identity relationship function shall enable entities to connect with each other and establish trust-based relationships that can be used to facilitate secure and interoperable communication and exchange of information. For example, an entity might use the identity relationship function to establish a trusted connection with another entity in another metaverse application, to exchange information or access resources.
- b) The identity relationship function shall define and provide supporting to manage the data, including identity data, or interaction result data, when the data is generated in any interaction scenarios inside an identity relationship in different metaverse applications. Especially in metaverse systems, entities like smart devices, AIs/robots, or sensors may be able to manage their data, while the data may be generated with in an identity relationship. For example, when driving a car, the driving data shall belong to the identity relationship between the car and the driver.
- c) The identity relationship function may include features such as the ability to establish, manage, and terminate relationships, as well as to manage and store information about the relationships. It may also include the ability to verify and authenticate the identities of entities involved in the relationships, and to verify that the relationships are secure and private.

The interoperable communication function of an identity framework shall enable different identity systems, applications, and services to communicate and exchange identity information in a standardized and interoperable manner. The goal of the interoperable communication function is to enable seamless and secure sharing of identity information between different systems, applications, and services in the metaverse. This includes support for the exchange of verifiable claims, the discovery of identity information from different sources, and the mediation of access to that information in a secure and privacy-preserving manner.

The interoperable communication function should verify that identity information can be shared and used in a way that is consistent, secure, and privacy-preserving across metaverse applications. This function is also key for enabling the creation of trusted relationships between identities, and for enabling the secure exchange of value and assets between different systems and services in multiple metaverses. The interoperable communication function shall meet the following requirements:

- The identity credential/claim management function shall be able to manage and verify identity credentials and claims in a secure and standardized manner.
 - a) Identity credentials and claims are digital representations of an entity's identity information, such as their name, address, age, qualifications, and other relevant details. These credentials and claims can be used to verify an identity of any entity in metaverses, as well as to provide access to certain resources and services.
 - b) The identity credential/claim management function shall be responsible for creating, storing, and managing identity credentials and claims, as well as for verifying their authenticity and validity. This function shall verify that identity information is secure and trustworthy, and that it can be used in a consistent and standardized manner across different systems, applications, and services.
 - c) The identity credential/claim management function should provide supporting to apply identity credentials or claims for all self-sovereign entities inside metaverse applications, such as smart devices, AIs/robots, and sensors.
 - d) The identity credential/claim management function may use secure cryptographic techniques, such as digital signatures and encryption, to help protect identity information, as well as to verify that it can be trusted by third-party entities. These techniques may include verifiable credential (VC) or anonymous credential (AC).

- The identity linked data function should be able to manage and connect identity information as linked data. Standard web protocols, such as HTTP and RDF, can be used in this function. (Linked data is a way of publishing and connecting information on the web in a standardized, machine-readable format. By linking data in this way, it becomes possible to create a web of interconnected information that can be easily shared, reused, and connected with other data sources.)
 - a) In the context of an identity framework, the identity linked data function should be responsible for representing and managing identity information, from multiple metaverse applications, as linked data. This allows identity information to be shared, reused, and connected with other data sources, such as social media profiles, online repositories, metaverse applications, and other identity systems.
 - b) The identity linked data function may support the discovery of identity information from different sources, as well as the mediation of access to that information in a secure and privacy-preserving manner.

5.5 Requirements for identity tool and maintenance functions for metaverse identity framework

The identity tool and maintenance function of an identity framework may define and provide supporting to different functionalities that support and enhance the management of digital identities in the metaverse, and support more identity based scenarios in metaverses. These tools should be designed with a focus on user experience so that users can easily use them and access the benefits that they offer in a seamless and secure manner. These tools may include functions such as identity presentation, which enables individuals or entities to display and use their identities in various contexts; identity value estimation, which helps determine the value or trustworthiness of an identity; and security and regulation, which protects the privacy and security of identity data, in compliance with relevant regulations and standards. The identity tool and maintenance function shall meet the following requirements:

- The identity presentation tools help users present their digital identity in a way that is consistent, trustworthy, and secure. This could be achieved through the use of digital certificates, visual representations, or digital signatures.
 - a) The identity presentation tool helps present that the identity information in a consistent and trustworthy manner, enabling the individual or entity to more effectively prove their identity and participate in metaverse activities.
 - b) To support one identity cross multiple metaverse scenarios, the identity presentation tool shall handle identity presentation based on different identity profiling that fit for each metaverse.
- The identity value estimation tool shall provide the ability to determine the value or worth of an identity within the metaverse. This tool helps users understand the value of their digital identity, such as the number of connections they have in a metaverse, their reputation, or other attributes. This tool can help users make informed decisions about how to manage and protect their digital identity.
 - a) The identity value estimation tool should enable individuals or entities to collect their identity resources, quantify the strength and reputation of their identity, and assess its value in different use scenarios.
 - b) The identity value estimation tool should consider factors such as the accuracy, completeness, and verifiability of identity information; the number and type of verifiable claims; and the level of trust of the identity issuers.
 - c) The identity value estimation function should provide an important feedback mechanism for individuals and entities to understand the value of their identity and take appropriate steps to improve it. It is a key component that helps society build-up for a metaverse.

- The security and regulation tool shall be responsible for protecting digital identities against cyber-attacks and for compliance with data protection and privacy regulations. The use of identity information within the metaverse shall be secure, trustworthy, and in compliance with all applicable legal requirements.
 - a) The security and regulation tool shall include measures to help reduce unauthorized access to or use of identity information, such as encryption and access controls. It also includes mechanisms for detecting and responding to security incidents, such as data breaches.
 - b) Data privacy and security are highly complex and increasingly regulated areas of law, and no standard can provide unconditional consistency with all applicable laws and regulations, which may also vary at the local, state, and regional level. The security and regulation tool should evaluate any implementation for considerations of data privacy, security, and data ownership in the context of an identity framework for metaverse and are responsible for conformance with all such laws and regulations.
 - c) Additionally, the security and regulation tool may help verify that the identity framework complies with relevant laws and regulations, including all applicable privacy laws, data protection regulations, and cybersecurity standards.
- There may be other useful identity tools that serves the metaverse better.

Annex A

(informative)

Bibliography

Bibliographical references are resources that provide additional or helpful material but do not need to be understood or used to implement this standard. Reference to these resources is made for informational use only.

[B1] Identifiers, D., (DIDs) v1.0.¹²

[B2] “The Road to the Metaverse: Grand Challenges and Roadmaps.” Dr. Yu Yuan. IEEE Metaverse Congress.¹³

[B3] Verifiable Credentials Data Model v1.1.¹⁴

¹²Available at <https://www.w3.org/TR/did-core/>.

¹³Available at <https://metaverse.ieee.org/>.

¹⁴Available at <https://www.w3.org/TR/vc-data-model/>.

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